RIVER VALLEY HIGH SCHOOL
YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE NAME	
CLASS	6
CENTRE NUMBER	S
INDEX NUMBER	

H1 CHEMISTRY 8872/02

Paper 2

24 September 2013

2 hours

Candidates answer Section A on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, class, Centre number and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** the questions.

Section B
Answer **two** questions on separate answer paper.

A Data Booklet is provided. Do not write anything on it.

The number of marks is given in brackets [ ] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
B5	
B6	
B7	
Total	

This document consists of **13** printed pages

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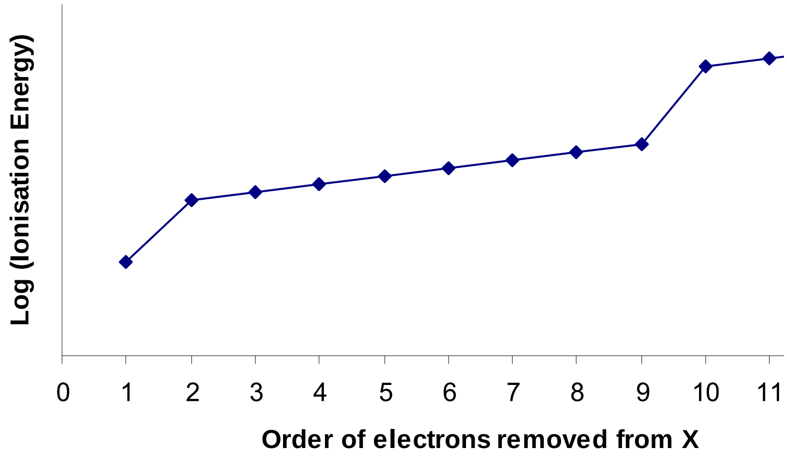
Section A [40 marks]

Answer **all** questions in this section.

1 **(a)** **X** is an element in the Periodic Table. The ionisation energies of the first 11 electrons in element **X** is plotted against the order of removal of electrons as shown in the diagram below:



(i) Suggest why log (ionisation energy) used as the vertical axis, instead of ionisation energy?

To reduce the scale on the vertical axis.

(ii) Account for the shape of the graph as observed.

There is a general increasing trend in successive ionization energies as the electron is being removed from ions of increasingly positive charge.

There is a sharp increase in ionization energy between the first and second electron removal as well as between the 9th and 10th electron removal as they involve the removal of electron from the inner principal quantum shell.

(iii) Identify **X**. Hence, state the formula of the oxide of **X**. Explain how you arrive at your answer.

X is sodium.

Na_2O.

The sharp increase in the 2nd ionisation energy indicates that the **<u>2nd electron is removed from an inner shell</u>**. Hence, the outer shell of an atom of **X** contains 1 electron. **X** is from Group **I** of the Periodic Table. [5]

(b) Give the oxidation state of sulfur in SO_3^{2-}.

+4 [1]

(c) Write equations for the reactions of each of these chlorides with water. What is the effect of adding chlorides to water?

(i) sodium chloride *(include state symbols in your equation)*

$NaCl(s) \rightarrow Na^{+}(aq) + Cl^{-}(aq)$

Dissolves to give a neutral solution (pH = 7)

(ii) silicon (IV) chloride

$SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl$

Dissolves and undergoes hydrolysis to form a strongly acidic solution

(iii) phosphorus (V) chloride

$PCl_5 + H_2O \rightarrow H_3PO_4 + 5HCl$

Dissolves and undergoes hydrolysis to form a strongly acidic solution [6]

[Total: 12]

2 (a) Explain what is meant by the half-life of a reaction.

Half-life is the **time taken for the concentration of a reactant to be halved**. [1]

(b) The radioactive decay of uranium-238 to radon-222 is a first-order process. The half-life of uranium-238 is 1630 years. How long will it take for a sample of uranium-238 to decay to 20% of it's original radioactivity?

Let number of half-lives be a. [2]

$$\left(\frac{1}{2}\right)^{a} = \frac{20}{100}$$

$$a\ln\frac{1}{2} = \ln\frac{20}{100}$$

a = 2.32

Time taken = 3.32 × 1630 = **3780 years (3 s.f.)**

OR

$$\ln\frac{20}{100} = -kt$$

Since $k = \frac{\ln 2}{t_{½}}$,

ln 0.2 = -t (ln 2 / 1630)

t = 3780 years (3 s.f.)

(c) In aqueous solution, iodic acid reacts with potassium hydroxide as such:

$$HIO_3 + KOH \rightarrow KIO_3 + H_2O$$

The reaction is carried out three times using varying concentrations of the reactants. The initial rate of the reactions were determined and the results are shown in the table below:

Experiment	$[HIO_3]$ / mol dm^{-3}	[KOH] / mol dm^{-3}	Initial Rate / mol $dm^{-3}s^{-1}$
1	0.040	0.030	3.25×10^{-4}
2	0.120	0.030	2.93×10^{-3}
3	0.040	0.080	8.66×10^{-4}

(i) Using the data given in the table, deduce and explain the order of reaction with respect to each reactant.

Comparing experiments 1 and 2,

When $[HIO_3]$ increases by a factor of 3 with [KOH] constant, the rate of reaction increases by a factor of 9.

Hence, order of reaction with respect to HIO_3 is two.

Comparing experiments 1 and 3,

When [KOH] increases by a factor of 2.7 with $[HIO_3]$ constant, the rate of reaction increases by a factor of 2.7.

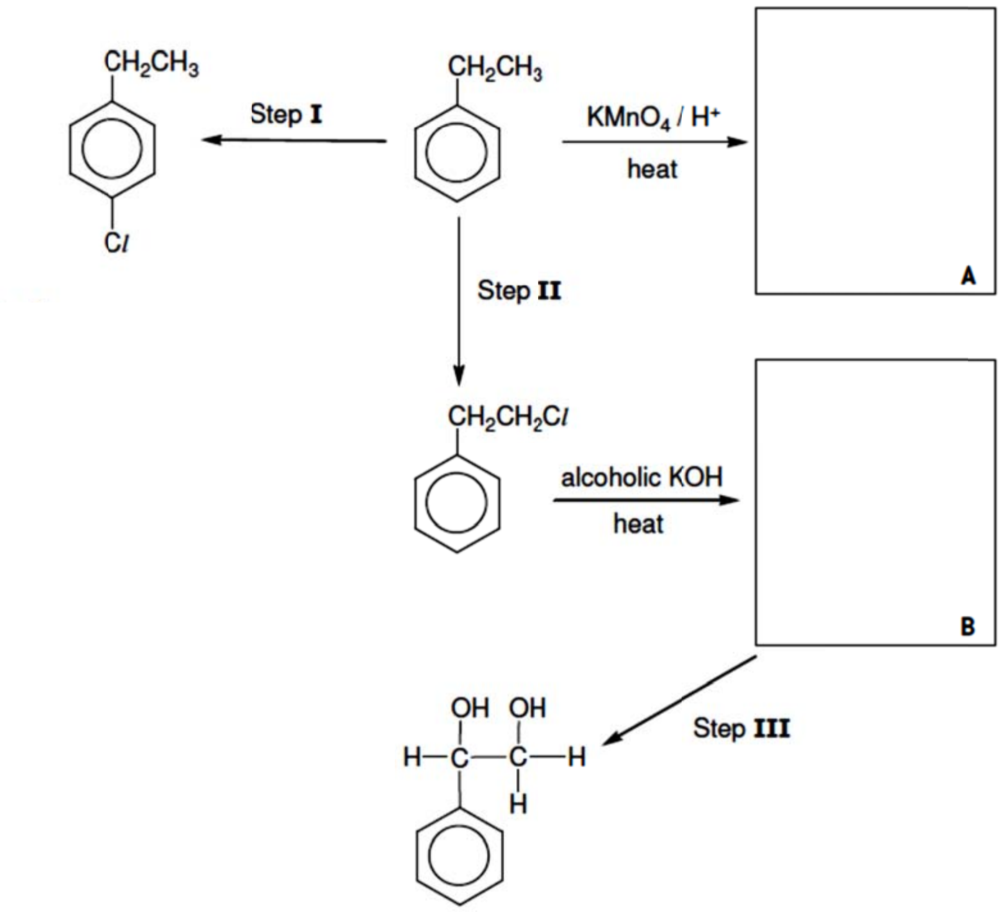
Hence, order of reaction with respect to KOH is one.

(ii) Hence, write a rate equation for the reaction.

Rate = $k[HIO_3]^2[KOH]$ [3]

[Total: 6]

3 **(a)**



Compound X

The flowchart above shows the various reactions of ethylbenzene. [7]

(i) Suggest the displayed formula of compound **A** and compound **B**.

A

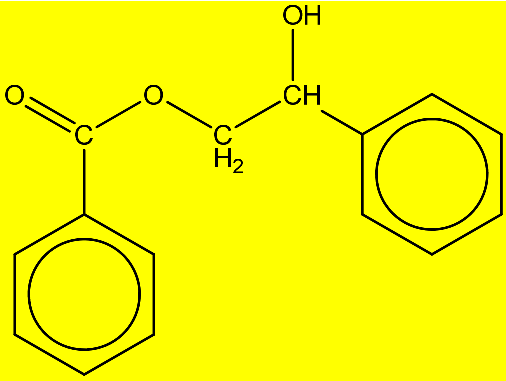
(ii) State the reagents and conditions for Steps **I, II** and **III**.

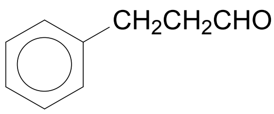
Step I:	$FeCl_3$, Cl_2, heat
Step II:	Limted Cl_2 u.v light
Step III:	Cold dilute alkaline $KMnO_4$

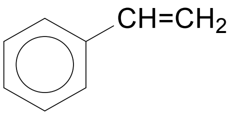
(iii) State the type of reaction in step **I**.

(Electrophilic) substitution

(iv) Draw the compound formed when limited amount of Compound **A** reacts with compound **X** in the presence of hot concentrated sulphuric acid under reflux conditions.



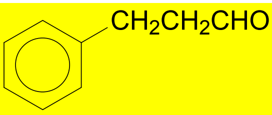
(b) Suggest how you would distinguish between 

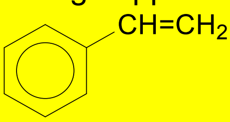
and 

using suitable chemical tests. State clearly the reagents and conditions used, equations as well as the observations for each of the following compounds. [3]

Test 1

2,4 DNPH

Orange ppt seen in 

, no precipitate observed in 

Eqn

Br_2(aq, $Br_2(CCl_4)$, tollen's reagent, fehling's reagent, are accepted too.

[Total : 10]

Use of the Data Booklet is relevant to this question.

4 Compressed natural gas (CNG) is a fossil fuel substitute for petrol, diesel fuel and propane. It is widely considered as a more environmentally friendly alternative to conventional fuels. In a world where fossil fuels are being depleted, CNG is rapidly gaining popularity as an energy source.

CNG is made by compressing natural gas, which is mainly composed of methane, CH_4, to less than 1% of the volume it occupies at standard atmospheric pressure. It is stored and transported in cylindrical containers. The mass of a full tank of CNG is approximately 18 kg. It is used in automobiles where combustion of CNG in the internal engine releases energy that in turn drives the automobile.

However in recent years, car companies have been adopting the option of using liquefied natural gas (LNG) in cylindrical containers of the same size as CNG tanks as an alternative in favour of CNG.

CNG's specific energy density is measured to be 42% that of (LNG).

Specific energy density is the amount of energy stored in a given system per unit mass. It is measured in mega joules per unit mass (MJ kg^{-1}) where 1 megajoule = 1×10^6 J.

(a) **(i)** Define the term *enthalpy change of combustion*.

The enthalpy change of combustion of a substance is the enthalpy change when **one mole of the substance is *completely* burnt in oxygen** [1]

(ii) Find the amount of energy released from one cylinder of CNG given that $\Delta H_c(CH_4)$ = -891 kJ mol^{-1}. Express your answer in terms of MJ.

Amount of methane = 18000 / 16 = 1125 mol

Amount of energy released = 1125 x 891 x 10^{-3} = 1002 MJ [2]

(iii) Hence, find the specific energy density of CNG.

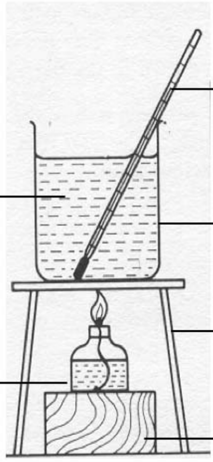
Specific energy density of CNG = 1002 / 18 = 55.7 MJ kg^{-1} [1]

(iv) Calculate the specific energy density of LNG? Explain, with reference to the difference in specific energy density values of CNG and LNG, why car companies prefer the use of LNG over CNG in recent years.

Specific energy density of LNG = 55.7 / 42.0 x 100 = 133 MJ kg^{-1}

The specific energy density of LNG is more than that of CNG which implies more energy can be released per unit mass of LNG. Hence, LNG is more space and energy efficient and is preferred over CNG. [2]

(b) Besides methane, many other organic compounds are also used as an energy source. An example is the usage of methanol in spirit lamps for the heating of water as shown in the diagram.



The volume of water in the beaker was 250 ml. Its initial temperature was 25.1 ºC and the highest temperature reached after heating was 28.6 ºC. The mass of the spirit lamp with methanol was originally measured to be 51.6 g. After heating, the mass was 51.4 g.

(1 ml = 1cm^3)

The density of water is 1 g cm^{-3}.

(i) Calculate the enthalpy change of combustion for methanol in kJ mol^{-1}.

Mass of methanol used in combustion = 51.6 – 51.4 = 0.2g

Amount of methanol = 0.2 / 32.0 = 0.00623 mol

Amount of energy absorbed by water = 250 x 4.18 x (28.6 – 25.1) = 3657.5 J = 3.66 kJ

Hence, $\Delta H_c(CH_3OH)$ = 3.66 / 0.00623 = -587 kJmol^{-1} [3]

(ii) The theoretical enthalpy change of combustion of methanol is -715 kJ mol^{-1}. Comment on the difference between the calculated and theoretical values.

The enthalpy change of combustion calculated in **b(ii)** is less negative than the theoretical value as heat was lost to the surroundings. [1]

(c) Methanol is an alcohol. Another alcohol that is frequently used as an energy source is ethanol. Apart from combustion, ethanol can also undergo oxidation to form ethanoic acid when reacted with hot acidified potassium manganite (VII).

Write an overall balanced equation to represent the reaction.

$CH_3CH_2OH + H_2O \rightarrow CH_3COOH + 4H^+ + 4e$

Overall:

[2]

$$5CH_3CH_2OH + 4MnO_4^- + 12H^+ \rightarrow 5CH_3COOH + 4Mn^{2+} + 11H_2O$$

[Total: 12]

Section B [40 marks]

Answer **two** of the three questions in this section on separate answer paper.

5 There are many different organic compounds that are commonly used in our daily life. For example, ethanoic acid is also common known as vinegar. Halogen derivatives are organic compounds which contain at least a C-X bond where X can be C*l*, Br or I. Halogen derivatives undergoes many reactions to form other compounds.

(a) In a laboratory experiment, a student reacted chloropropane, bromopropane and iodopropane with $Ag(NO_3)$ in the presence of aqueous NaOH under suitable conditions to produce silver halide precipitate. **[4]**

(i) State the colour of the precipitates formed from the above experiment.

AgC*l* --- White ppt, AgBr--- Cream ppt, AgI--- Yellow ppt

(ii) It is also observed that the rate of the formation of the precipitates differs.

With the help of data booklet, state and explain the rate of formation of the three precipitates.

The rate of formation for AgI is the fastest while formation of AgC*l* is the slowest.

(The alkyl halides undergo nucleophilic substitution with H_2O which involves breaking of the C–X bond and releasing the halide ion.

The halide ion formed gives a precipitate with $AgNO_3$(aq).)

The C–X bond strength increases in the order *E*(C–I) < *E*(C–Br) < *E*(C–C*l*)

E(C–I) = 240 kJ mol^{-1}

E(C–Br) = 280 kJ mol^{-1}

E(C–C*l*) = 340 kJ mol^{-1}

From the values, it is shown that C-C*l* has the highest bond energy while C-I has the lowest bond energy. Hence, the C–I bond breaks most easily to give AgI ppt most quickly, followed by C–Br bond to give AgBr ppt and then C–C*l* bond to give AgC*l* ppt most slowly.

(b) Organic acids and organic alcohols are known to be acidic. The K_a values of some organic acids and alcohols are shown below.

Type of organic compound	K_a values /mol dm^{-3}
Propanoic acid (CH_3CH_2COOH)	1.34×10^{-5}
2-chloropropanoic acid ($CH_3CHClCOOH$)	1.48×10^{-3}
Propanol ($CH_3CH_2CH_2OH$)	1.00×10^{-16}

Explain the above observations. **[3]**

Acid Strength : Propanol <propanoic acid < 2-chloropropanoic acid

Propanoate anion is more stable than the $CH_3CH_2CH_2O^-$ as the negative charge is delocalised over the COO^- group. On the other hand, the electron-donating ethyl group intensifies the negative charge on oxygen of $CH_3CH_2CH_2O^-$ anion.

2-chloropropanoic acid is a stronger acid than propanoic acid as the presence of electron-withdrawing *Cl* atom disperses the negative charge on the COO^- group, making the substituted carboxylate anions more stable than $CH_3CH_2COO^-$.

(c) Many fruit flavours and aromas that are used in the food industry are esters. An example of the esters is ethyl propanoate which is used as a kiwi or strawberry flavours. **[3]**

(i) Write an equation to show the hydrolysis of ethyl propanoate in basic conditions.

$CH_3CH_2COOCH_2CH_3 + NaOH \rightarrow CH_3CH_2COO^-Na^+ + CH_3CH_2OH$

(ii) Briefly outline how you would produce ethyl propanoate in a school laboratory.

Add Ethanol and propanoic acid into a flask. Add a few drops of concentrated sulfuric acid as catalyst heat/ heat under reflux.

(d) Ethanoic acid has many uses and can undergo many different reactions. **[10]**

(i) $CH_3OH(l) + CH_3CO_2H(l) \rightleftharpoons CH_3CO_2CH_3(l) + H_2O(l)$

1. Define “dynamic equilibrium”

Dynamic equilibrium refers to a reversible process at equilibrium in which the rate of the forward reaction is equal to the rate of the backward reaction.

2. Calculate the K_c for the above reaction at equilibrium given that 3 dm^3 of 2.00 mol dm^{-3} of ethanoic acid and 3 dm^3 of 2.00 mol dm^{-3} methanol is used and the degree of dissociation is 0.15

	$CH_3OH(l)$ +	$CH_3CO_2H(l)$ ⇌	$CH_3CO_2CH_3(l)$ +	$H_2O(l)$
Initial conc /mol dm^{-3}	0.33	0.33	0	0
ΔC /mol dm^{-3}	–0.0495	–0.15	+0.15	+0.15
Eqm conc /mol dm^{-3}	0.281	0.281	0.2	0.2

$K_c = [CH_3CO_2CH_3][H_2O] / [CH_3OH][CH_3CO_2H]$

$= (0.20)^2 / (0.281)^2 = 0.506$

(ii) In a separate experiment, 25.0cm^3 of 0.50 mol dm^{-3}ethanoic acid is titrated with 0.35 mol dm^{-3} sodium hydroxide.

1. Sketch a titration curve for the above reaction, indicating the buffer region and volume at equivalence point.

- Correct shape of graph
- Correct indication of buffer region(before equivalence point)
- Volume at equivalence point (35.70 cm^3)

2. Suggest a suitable indicator for this titration

Phenolphthalein, bromothymol blue, thymol blue

3. *"Ethanoic acid behaves like a strong acid in an acid-base titration."* Explain, with the help of equation(s), this statement.

$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$

$H^+ + OH^- \rightarrow H_2O$

When OH^- is added, it will react with H^+ present. Hence the equilibrium position will shift to the right to replenish the H^+. At the end of the titration, the ethanoic acid has fully dissociated. Hence ethanoic acid behaves like a strong acid in the titration.

[Total: 20]

6 **(a)** A reaction can occur between sodium thiosulfate and hydrochloric acid. The reaction is represented by the equation below:

$$Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow S(s) + SO_2(g) + 2NaCl(aq)$$

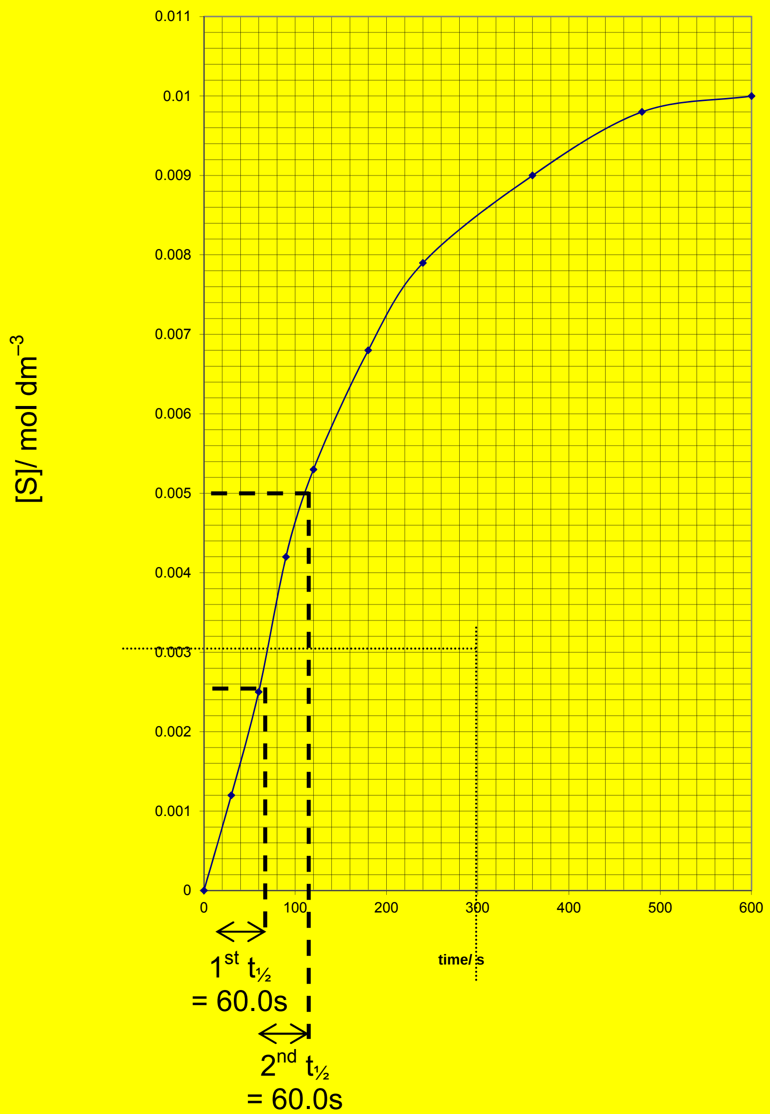
The kinetics of this reaction was investigated. The initial concentration of hydrochloric acid was 0.100 mol dm^{-3} which is in excess. Hence the rate of reaction is assumed to be dependent only on the concentration of $Na_2S_2O_3$(aq). The reaction was monitored using a photometer, an instrument which measures the intensity of the yellow sulfur formed at regular time intervals. The measurements were then compared against those obtained from samples containing known concentrations of sulfur.

The results obtained are summarised in this table:

Time/s	0	60	120	180	240	480	600
[S]/ mol dm^{-3}	0	0.0025	0.0053	0.0068	0.0079	0.0098	0.0100

(i) Plot the graph of [S] against time on the graph paper provided Use your graph to determine the order of the reaction with respect to $Na_2S_2O_3$(aq).

Show clearly how you arrived at your answer.



Graph with correct plotting:
Half-life clearly indicated:
Correct axes, labels:
Since half-lives are constant at 60.0s,
Order of reaction with respect to $Na_2S_2O_3$(aq) is one.

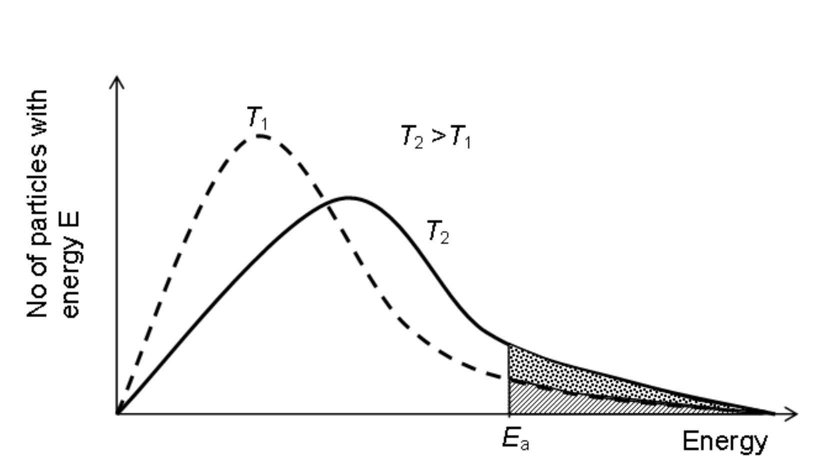
(ii) Given that the order of this reaction is 1 with respect to hydrochloric acid, and the rate of the reaction is 2.25 x 10^{-4} mol dm^{-3} s^{-1} when [$Na_2S_2O_3$] = 0.0035 mol dm^{-3}, give the rate equation for the reaction and calculate the rate constant, stating its units. [7]

Rate = k[$Na_2S_2O_3$][HC*l*]

Rate = 2.25 x 10^{-4} mol dm^{-3} s^{-1} and [$Na_2S_2O_3$] = 0.00355 mol dm^{-3}
Substitute into the rate equation: 2.25 x 10^{-4} = k x 0.1 x 0.00355.

Solving, k = 2.25 x 10^{-4}/(0.1 x 0.00355) = 0.634 mol^{-1} dm^{3} s^{-1}

(b) The reaction between sodium thiosulfate and hydrochloric acid was [3] carried out at standard conditions. With the aid of a Boltzmann distribution curve, illustrate how an increase in temperature can increase the rate of the reaction.



Peak moves to the right:

Peak of higher T curve lower than peak of lower T curve:

Increase in number of particles with energy greater than or equal to E_a: When temperature increases, the average kinetic energy of the reactant particles increases . As a result, the frequency of collisions between reactant particles increases and the rate of reaction increases .

(c) Sodium, sulfur and chlorine are elements in Period 3. With the aid of a diagram, explain the variation in electrical conductivity of Period 3 elements. [8]

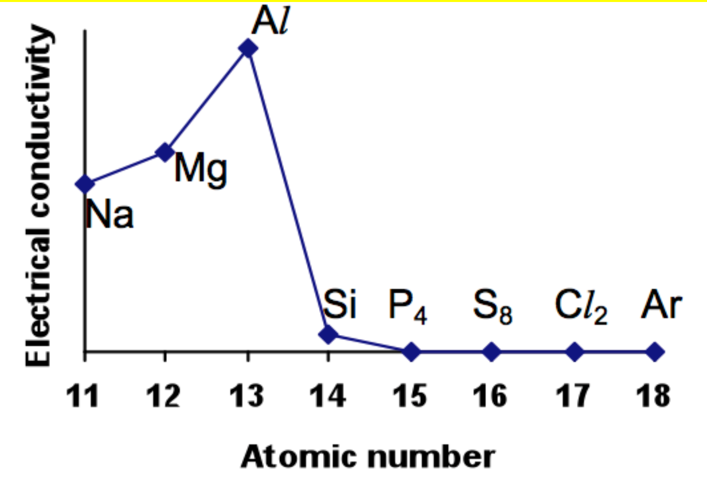


Diagram: Correct shape and correct axes (1m each)

Electrical conductivities are high for the metals Na, Mg and A*l*, and increases from Na to A*l*. Na, Mg and A*l* have giant metallic structures and the number of delocalised valence electrons increases from Na to A*l*.

Electrical conductivity drops sharply at Si. Si has a giant covalent structure and is a semi-conductor.

Electrical conductivities drop to zero at P_4 and remains at zero to Ar. P_4 to

Cl_2 have simple covalent structures and Ar has a monoatomic structure and there are no mobile charge carriers to conduct electricity.

(d) When aluminium chloride is dissolved in water, the resultant solution turns blue litmus paper red. Write equation(s) to show why this is so. [2]

$AlCl_3(s) + 6H_2O(l) \rightarrow [Al(H_2O)_6]^{3+}(aq) + 3Cl^-(aq)$
$[Al(H_2O)_6]^{3+}(aq) + H_2O(l) \rightleftharpoons [Al(H_2O)_5(OH)]^{2+}(aq) + H_3O^+(aq)$

[Total: 20]

7. **(a)** This question is about compound **A**, $C_{14}H_{18}O_3$.

A reacts with 2,4-dinitrophenylhydrazine to give an orange precipitate but no reaction with Fehling's reagent.

However, silver mirror is observed when **A** is added to hot silver diammine complex solution. **B** and **C** are produced from this reaction.

Acidification of **B** produce **D** with molecular formula $C_{10}H_{10}O_4$.

D reacts with Na metal. Effervescence is observed. 1 mole of hydrogen gas is produced. **B** is produced again. **D** shows no reaction with acidified $KMnO_4$ while **E** decolourise it.

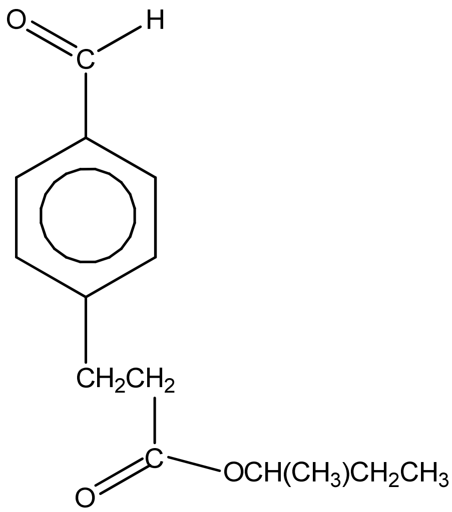
Pale yellow precipitate with an antiseptic smell is observed when **C** reacts with a solution of aqueous iodine and sodium hydroxide.

Identify compounds, **A** to **D** by providing their structural formulae. Explain the chemistry of the reactions described. [10]

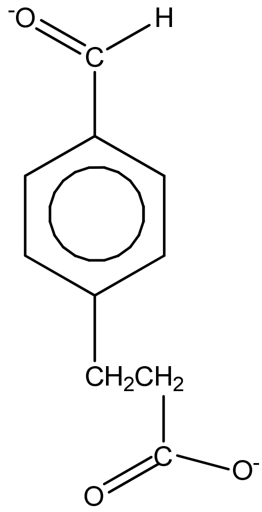
- A has a benzene ring as C:H ratio is 1:1
- **A** undergoes condensation with 2,4-DNPH $\Rightarrow$ **A** is a carbonyl compound/ketone/aldehyde.
- **A** undergoes oxidation with tollen's reagent but not fehling's reagent $\Rightarrow$ **A** is a benzaldehyde
- **A** undergoes alkaline hydrolysis to form **B** and **C**
  $\Rightarrow$ **A** contains an ester.
- **D** undergoes acid-metal reaction to produce 1 mole of H_2 gas $\Rightarrow$ **D** is has 2 carboxylic group.
- **E** undergoes oxidation with acidified $KMnO_4$ $\Rightarrow$ **E** is an alcohol.
- **E** undergoes oxidation with alkaline I_2 $\Rightarrow$ E contains a $CH(CH_3)(OH)$ group.

A:

Mark awarded for substitution at different position.



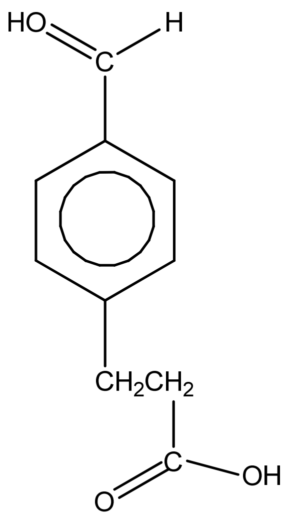
B:



C:

$CH_3CH_2CH(CH_3)OH$

D:



(b) Phosphorous and carbon are non-metals found in the periodic table. Many derivatives of carbon and phosphorous are found to be useful and common in chemistry

Benzene is an organic chemical compound with the formula of C_6H_6. Benzene is a colourless and flammable liquid with a sweet smell and relatively high melting point. Many important chemicals are derived from benzene where one or more hydrogen atoms is replaced with another functional group. **[4]**

(b) **(i)** Explain the formation of the shape of benzene in terms of σ and π bonds.

In benzene, three sp^2 orbitals and one unhybridised 2p orbital is formed.
Two sp^2 orbitals of the C atom overlap head-on with the sp^2 orbitals of two neighbouring C atoms, forming two C—C σ bonds.
The sp^2 orbital of the C atom overlaps head-on with the 1s orbital of one

H atom, forming a C—H σ bond.

The unhybridised 2p orbitals of all the six C atoms overlap sideways to form a delocalised π electron system/cloud above and below the plane of C atoms.

(ii) State the hybridisation for the carbon atoms in benzene.

sp^2

(c) PCl_3 and PCl_5 are two common reagents used in organic synthesis.

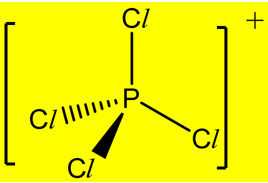
PCl_5 is soluble in polar solvents. Previously, PCl_5 in solution was thought to form a dimeric structure, P_2Cl_{10}. In more recent times, it has been proven that PCl_5 is soluble in a polar solvents in the "ionic" form of P_2Cl_{10} as shown in the equation below:

$2\ PCl_5 = [PCl_4^+][PCl_6^-]$

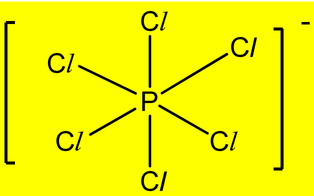
State and draw the shapes of the ions of PCl_4^+ and PCl_6^- and suggest a reason why PCl_5 is said to be soluble in its "ionic" form. **[6]**

(i) State and draw the shapes of the ions of PCl_4^+ and PCl_6^-

Shape of PCl_4^+ : tetrahedral



Shape of PCl_6^- : octahedral



(ii) Suggest a reason why PCl_5 is said to be soluble in its "ionic" form.

The "ionic" form of P_2Cl_{10} is soluble in polar solvents via ion-dipole interactions.

(iii) Phosphorous can exist as PCl_3 and PCl_5. However, nitrogen cannot exist as NCl_5. Explain this observation.

Nitrogen is in period II. It is unable to expand its octet structure as it does not have accessible d orbitals available.

[Total: 20]

END OF PAPER